

What this adds

This is the standard NanoEls H4 firmware with a good deal added. Everything the original does is still here and behaves the same way — the list below is what is new. Roughly half the additions are about setting the machine up and checking it is right, rather than about cutting metal.

Set it up without a computer		Easier to read	
Settings on the panel	The stock firmware is configured by editing the source and reflashing it. Here every figure — screw pitches, motor steps, speeds, backlash — is a menu item on the machine.	Big position display	Z and X in digits you can read from a few feet away, instead of squinting at the panel.
It measures itself	Fourteen guided routines work out the numbers for you on the actual lathe: how far each screw really moves, how much slack it has, how fast the motors can go before they stall, and how finely the spindle sensor reads.	Diameter readout	Shows the cross slide as a diameter, which is how lathe work is measured, rather than as distance from centre.
Save and restore	One command over USB prints every setting as text you can keep. Paste it back to restore a machine, or to set up a second one the same way.	Cutting speed helper	Tell it what you are cutting and with what, and it works out the spindle speed the current diameter wants — and keeps telling you as the diameter changes.
From a phone	The controller can make its own WiFi network, so you can change settings and update the firmware from a phone standing at the lathe. Off unless you switch it on.	Spindle indexing	Divides a turn of the spindle into equal marks and beeps as you reach each one. With slotting, that gives you hexagons and keyway sets without an indexer.
New things it can cut		A more trustworthy spindle sensor	
Powered cross-feed	The gearbox driving the cross slide instead of the carriage, so facing cuts are fed evenly rather than by hand.	Twice the resolution	Reads the sensor four times per pulse instead of twice, so it knows the spindle angle twice as precisely from the same encoder.
Slotting	Uses the lathe as a shaper. The spindle stays still and the tool strokes back and forth — keyways, flats and squares without a milling machine.	Sorted out in the menu	Counting direction, belt ratios and noise filtering are all settings now, rather than wires to swap or numbers to work out by hand.
Tapered threading	Threads cut on a taper, for pipe fittings such as NPT and BSPT.	It tells you when the wiring is bad	A live figure for how clean the signal is. Electrical interference from the motors is the usual cause of a thread going wrong, and nothing else on the machine can see it — a sensor twitching in place looks like a stopped spindle to everything else.
Thread list	52 common threads built in — metric, UNC/UNF, BSPP, trapezoidal, ACME and NPT. Choose one and the feed is set for you.	Small things that help	
Better cuts		Settings per job	How many passes, how much clearance and so on are remembered separately for each kind of job, so setting up a thread does not disturb your turning settings.
Spring passes	Repeats the last pass without going deeper, to take out the spring in the tool and get the size you actually asked for.	Numbers typed as you would write them	4mm is one keystroke, not four. The stock firmware takes every distance in whole microns, so the smallest useful figure sets the length of every figure; here the backspace key is a decimal point, and it refuses one where a fraction cannot be stored.
Angled thread infeed	Feeds the tool in at an angle so it cuts mainly on one side of the thread instead of both at once. Easier on the tool and leaves a better finish.	Sounds that mean something	A refused key, a finished pass and a lost thread each sound different, so you can keep your eyes on the work.
Peck parting	Backs the tool out every so often while parting off, to clear the swarf before it packs in and jams.	Optional joystick	A stick for jogging the axes, if you would rather not use the keypad.
One-key retract	Pulls the tool clear of the work and puts it back on exactly the same number.	Fewer ways to get it wrong	Four different things can be wired to the same six spare terminals; switching on one that would clash with another is refused and says which.

If you are already running the standard firmware

Your settings survive the update and the machine will behave as it did. Five things are worth doing once afterwards.

1	Save your settings first. Send \$ over USB and keep the text somewhere — there is no undo.
2	Leave the encoder PPR figure alone. It still means the same thing; the controller simply reads the sensor more finely and works the rest out itself.
3	Check the glitch filter . It is a noise filter, and because the sensor is now read twice as often, a setting that was safe before can start losing pulses at high speed. There is a formula in the manual.
4	Run Encoder signal from the calibration menu during a real cut and see what it reads. It should sit at 100. Anything less is interference on the sensor wiring, which is worth fixing properly rather than filtering out.
5	Anything you used to change by editing the source is now in the settings menu. The source file only supplies the starting values for a controller that has never been set up.

Everything it does

Feeding the tool		What it will tell you	
Screw-cutting gearbox	Any feed or thread pitch, metric or imperial, without change gears.	Where the tool is	Millimetres or inches, radius or diameter.
Powered cross-feed	The same, driving the cross slide.	Big digits	Readable from across the shop.
Steady feed	A fixed speed with the spindle out of it altogether.	Spindle speed and angle	Plus the cutting speed at the tool.
Taper turning	The tool moves in as it travels along, at a ratio you set.	What speed to run	For the material and tool you have told it about.
Jogging	Arrow keys, with a step size down to a hundredth of a millimetre.	Indexing	Equal marks around a turn, with a beep at each.
Handwheels	Up to two, each assignable to whichever axis you like.	Sensor health	Whether the spindle sensor is being interfered with.
		How far through	Which pass of how many, with a bar.
Jobs it will do for you		Setting it up	
Turning and facing	To a size, in as many passes as you ask for.	Everything on the panel	No reflashing to change a number.
Parting and grooving	With optional pecking to clear the swarf.	Self-measurement	Fourteen routines that work out your lathe's figures for you.
Threading	Straight or tapered, single or multi-start, from a list of 52 threads.	Thread list	52 threads across six families.
Slotting	Keyways and flats, using the lathe as a shaper.	Backup and restore	One text file, over USB or from the browser.
Balls and dishes	Rounded ends, convex or concave.	From a phone	The same settings, easier to type, with live readouts.
G-code	Programs written on a computer and sent over.	Firmware updates	Over WiFi, with no computer at the machine.
A fourth axis	If you fit one — rotary or linear.	Keeping you out of trouble	
Getting the size right		Emergency stop	If it loses track of position, or a key is stuck at power-up.
Backlash taken up	Automatically, whenever an axis reverses.	It says why	A refused action explains itself on screen and sounds its own tone.
Limits	Four marks that define the cut and cannot be overrun.	Wiring clashes	Two devices cannot claim the same terminals.
Belt ratios	On any axis and on the spindle sensor.	No changes mid-cut	Settings will not change underneath a move in progress.
Spring passes	A repeat at size to take out tool spring.		
Angled thread infeed	Cuts on one flank rather than plunging on both.		

Buying hardware for a conversion

General guidance rather than a shopping list — the right answer depends on your lathe, and a motor that suits a small hobby machine will not move the carriage on a big one. Two things get less attention than they deserve: the sensor wiring, and whether your motors can be trusted not to lose steps.

Part	What to look for
Controller	The NanoEls H4 board: two stepper outputs, a spindle sensor input, six spare terminals and a 20×4 display. Boards and the circuit files come from the original project.
Spindle sensor	An incremental rotary encoder, 600 to 1000 pulses per revolution, with two output channels (A and B). Mount it straight on the spindle if you can, or on a 1:1 belt. Gearing it to spin faster than the spindle means more pulses to keep up with, and there is a speed ceiling — the manual has the sum. Higher resolution is not automatically better.
Sensor wiring	Shielded cable with twisted pairs, the shield earthed at one end only, run well away from the motor and VFD cables. This matters more than which encoder you buy. Interference here is the most common cause of a thread coming out wrong, which is why the firmware watches for it.
Stepper motors	Size them to the axis rather than the lathe. The carriage is heavy and wants the larger motor; the cross slide moves far less and needs less. Closed-loop motors cost more but remove the one fault the controller cannot detect: a step it asked for that never happened.
Drivers	400 to 800 steps per turn of the screw is plenty. Finer microstepping sounds better but costs torque and buys nothing the screw can actually deliver. If a motor runs hot standing still, turn off Hold when idle for that axis.
Lead screws	Ball screws remove nearly all the slack and are worth it on the cross slide first. Ordinary trapezoidal screws work fine — backlash is taken up automatically — but compensation cannot rescue a screw that is worn unevenly along its length. The backlash routine will tell you what you have.
Power supply	24 to 48 volts for the motors, rated for both running at once. The controller itself runs from 5 volts.

Run the calibration routines again after any mechanical change. The controller only knows what it has been told or has measured, and swapping a screw or a motor turns every stored figure back into a guess.